Abhishek Vinayak Pai

avpai4@illinois.edu | 1010 W. Green St., Urbana, IL 61801 | (217) 904 2753

EDUCATION

University of Illinois Urbana Champaign — Aug. 2024 – Present
MS Aerospace Engineering, GPA 3.75/4 — Urbana, IL

Manipal Institute of Technology, Manipal — Aug. 2020 – Jul. 2024
B.Tech Mechanical Engineering, Minor in Computational Mathematics, GPA 3.5/4 — Manipal, India

SELECTED PROJECTS

Autonomous SLAM and Object Detection in Unknown Environments — Mar. 2025 – May. 2025
University of Illinois Urbana Champaign, Course Project — Urbana, IL

- Developed custom LiDAR-based SLAM and PID wall-tracking for autonomous exploration on an F1-Tenth vehicle.
- Integrated real-time object detection with map-based location reporting for precise object localization.

Developing and Evaluating VLA+RL Models for Robotic Control — Jan. 2025 – May. 2025
University of Illinois Urbana Champaign, Course Project (ongoing research project) — Urbana, IL

- Applied concept probes and causal interventions to investigate failure modes in VLA models fine-tuned with RL.
- Analyzed model reliability and decision-making in robotic manipulation tasks using interpretability techniques.

Genetic Algorithm–Driven Wing Design for Fixed-Wing Micro UAVs — Mar. 2025 – May. 2025
University of Illinois Urbana Champaign, Course Project — Urbana, IL

- Designed an end-to-end wing optimization pipeline by coupling genetic algorithms with vortex lattice method simulations for micro-UAVs.
- Automated aerodynamic sizing and performance evaluation under multiple flight constraints.

Micro UAV for Extended-Range Survey and Object-Aware Mapping — Jan. 2023 – May. 2023
Manipal Institute of Technology, Funded Project — Manipal, KA

- Developed a custom micro-UAV platform with integrated FPV for long-range exploration.
- Implemented SLAM-based indoor mapping with onboard object detection to enable object-aware map generation.

SELECTED RESEARCH

Safe Landing Zone Detection for UAVs using Image Segmentation and Super Resolution — **MVA 2023**
Anagh Benjwal, **Abhishek Pai***, Aditya Vadduri*, Prajwal Uday** — Hamamatsu, Japan

- Novel approach for safe landing zone detection for UAVs, leveraging DL-based image segmentation with a custom dataset and integrating SuperResolution, resulting in up to 6.3% improvement in accuracy over SOTA methods.

Precise Payload Delivery via UAVs using Object Detection Algorithms — **AICECS 2023**
*Aditya Vadduri, Anagh Benjwal, **Abhishek Pai**, Elkan Quadros, Prajwal Uday* — Manipal, India

- Novel navigation method that improves the accuracy of conventional navigation methods by incorporating a deep-learning-based computer vision approach, achieving a 5 times increase in average horizontal precision over conventional GPS-based approaches.

WORK EXPERIENCE

Skyrobee Aerolabs LLP — Apr. 2023 – Jun. 2024
Co-Founder & Design Engineer — Pune, India

- Led the design and development of multi-rotor UAVs for security applications and established the company's legal structure and sales pipeline. Secured multiple defense contracts for FY 2023–24.

MAHE Centre for Aerospace Science and Technology (CAST) — Dec. 2023 – Jun. 2024
Research Assistant, Undergraduate Thesis — Manipal, India

- Researched hybrid airship aerodynamics and shape optimization under *Dr. Manikandan Murugaiah*, using CFD to generate aerodynamic data and developed optimization algorithms. Published in ***The Aeronautical Journal*** (2025).

Airbus ODC Jun. 2023 – Jul. 2023
Software Intern Bangalore, India

- Developed a software tool to evaluate knockdown factors for composite structures, enabling automated analysis of manufacturing defects to support engineering concessions and improved evaluation efficiency by about 70%.

MENTORSHIP & TEACHING

Autonomous Vehicle Design Head Jan. 2022 - Aug. 2023
AeroMIT, Technical Team Manipal, India

- Led the design and development of over 20 fixed-wing and multirotor UAVs across diverse missions. Mentored a subsystem team of 8 and contributed to award-winning international competition entries.
- Represented the team at expositions and innovation conclaves, showcasing experimental and competition builds to international researchers and officials.

Course Assistant Jan. 2023 - May. 2023
Dept. of Aeronautical Engineering, MIT Manipal Manipal, India

- Honorary position awarded to undergraduate student scoring the highest grade in the AE101 – Introduction to Aerospace Engineering open-elective course.
- Taught classes on fundamental aeronautical engineering concepts to a cohort of 60 students and restructured course syllabus and flow for more impactful learning.

Workshop Instructor Oct. 2023
IE Aerospace × AeroMIT, MIT Manipal Manipal, India

- Instructed a technical workshop on fixed-wing UAV design and aerodynamic analysis, engaging 50+ students across multiple departments.

ACHIEVEMENTS & SCHOLARSHIPS

- Placed **2nd** in design, **4th** overall globally at SAE-Lockheed Martin Aerodesign 2023, Fort Worth, TX.
- Placed **1st** in design, **2nd** in mission performance & **2nd** overall globally at SAE-Lockheed Martin Aerodesign 2022, Van Nuys, CA.
- Placed **1st** overall nationwide at the SAE India Aerothon 2022, Bangalore, India.
- Awarded the MAHE *Undergraduate Research Seed Grant* of $2500 to support independent project development and prototyping for drone projects.
- Awarded the *Manipal Merit Scholarship* in 2020 for undergraduate studies.

SKILLS

Programming: MATLAB, Python, C/C++, Bash
Libraries & Frameworks: PyTorch, Scikit-learn, OpenCV, YOLO, OpenAI Gym, ROS, Gazebo
Tools: Git, ANSYS Workbench, Solidworks, XFLR5, OpenVSP

KEY COURSES UNDERTAKEN

Robotics	Optimal Aerospace Systems, Formal Methods in Aerospace Robotics, Principles of Safe Autonomy, Artificial Neural Networks, Computational Systems Engineering
Math & Statistics	Statistical Learning, Time Series Analysis, Computational Linear Algebra, Probability & Design of Experiments, Metaheuristic & Multi-Objective Optimization

OTHER INTERESTS

- Indian classical vocalist and multi-instrumentalist with over 8 years of stage experience. Cleared 3 levels of ABGMV vocals examination with distinction.
- Skilled at RC flying of both fixed wing UAVs and multi-rotors.